CELLPILOT FEASIBILITY STUDY

Kitting station

Kitting station · recorded 2026-10-17 · 3 cameras · 0:35 · stored result · model gemini-4-argon · analysed Oct 17, 2026, 12:25 PM

A kitting station. The operator takes parts from three bins (A, B, C) and places one of each into the slots of a tray, orients the third part, compares the kit with a reference card, applies a label, and puts the finished kit into a bag on the out pile.

1 of 3 steps can be automated · 1 stays human · 1 needs a fixture

Engineer: 0 of 3 approved · 3 not yet signed off

Across three observed cycles, the operator repeatedly picks a compact rectangular wireless power bank from the tabletop, transfers and reorients it between both hands while visually presenting it toward the workstation cameras, and places it back onto a target spot on the table. Although the station was nominally described as a multi-bin kitting and labelling station, only the single power bank handling and inspection task takes place in the recording. Picking the flat power bank from the open tabletop requires a locating fixture or recessed nest so a two-finger parallel gripper can grasp its thin profile reliably, while placing the rigid 150 g unit onto the table can be automated directly with any of the catalogued 6-axis cobots such as the UR5e or xArm 6. The in-air bimanual transfer, rotation, and visual presentation step should remain with a human operator unless dedicated multi-angle vision cameras replace the manual inspection.

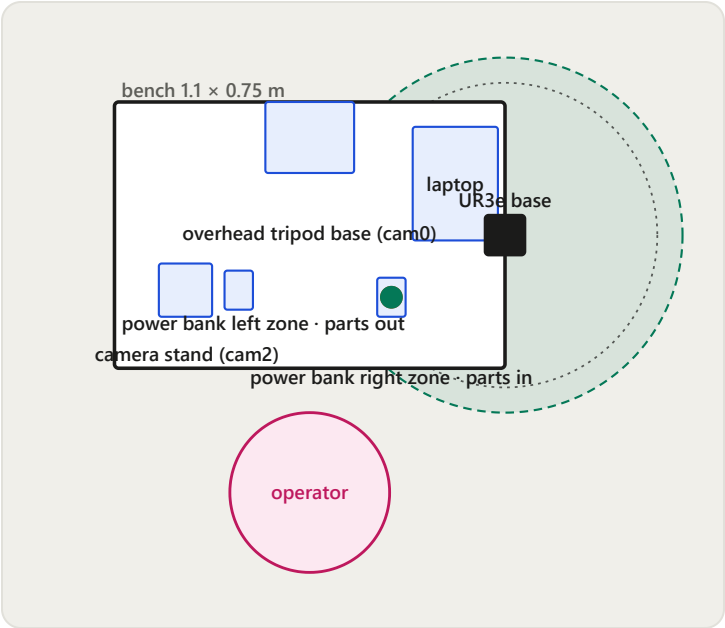
Steps, evidence and sign-off

Step and evidence	Verdict	Sign-off
1 · Pick power bank from table 3 occurrences · mean 1 s · confidence 0.9 • The power bank lies flat on the bare wooden tabletop without a bin or mechanical stop, so its pick coordinates shift between cycles and its thin 15 mm profile leaves minimal clearance above the table surface for parallel gripper jaws. `cam0 at 00:14.0` • The operator must scoop and pinch the thin edges of the unindexed power bank directly against the flat tabletop, which a cobot could perform reliably only if a raised nest or fixture presents the part in a known pose. `cam1 at 00:14.0` **Verifier:** All three pick occurrences match what is seen in cam0, cam1, and cam2 within one second, and picking the thin unindexed power bank off the flat tabletop requires a fixture. **Recommendation:** Picking the 15 mm thin power bank directly off the flat tabletop requires a raised nest or gravity presenter so parallel gripper jaws have clearance to grasp the edges without bottoming out. Fixture: Raised nesting block with lateral finger cutouts or a gravity-fed angled presenter.	NEEDS FIXTURE verifier agrees	not signed off

Step and evidence	Verdict	Sign-off
2 · Reorient and inspect power bank 3 occurrences · mean 1.9 s · confidence 0.9 • The operator performs an unscripted mid-air hand-to-hand transfer to flip and reorient the power bank, which cannot be executed by a single 6-axis arm with a two-finger parallel gripper without an intermediate regrip station. `cam0 at 00:15.0` • The operator angles and presents the face of the power bank toward the camera for visual inspection, requiring human visual judgement and dexterous wrist articulation. `cam2 at 00:16.0` **Verifier:** In cam0, cam1, and cam2 during cycle 2, after presenting the power bank toward cam2 at 16.0 s, the operator continues holding it in mid-air and transfers it back from his right hand to his left hand until 17.8 s rather than ending at 16.5 s. **Recommendation:** This step stays with the operator because dexterous hand-to-hand flipping and visual inspection cannot be done by a single arm without a secondary regrip station and an automated vision camera system.	KEEP HUMAN pre-flagged verifier timing differs 1.3 s (corrected)	not signed off
3 · Place power bank on table 3 occurrences · mean 1.2 s · confidence 0.88 • Placing the rigid rectangular power bank onto the flat open tabletop requires a simple vertical descent and gripper release with no tactile feedback. `cam0 at 00:22.0` • The target placement area on the table is unobstructed and well within the reach and payload capacity of a standard 6-axis cobot with a parallel jaw gripper. `cam2 at 00:21.0` **Verifier:** In cam0, cam1, and cam2, cycle 1's 8.8–14.0 s window encompasses an initial placement at 8.8–10.0 s followed by an in-air lift and inspection from 10.0–13.0 s before the actual placement at 13.0–14.0 s, and in cycle 2 the operator is still transferring the part between hands in mid-air until placement begins at 17.8 s. **Recommendation:** Automate with a standard two-finger parallel gripper using contoured urethane pads to descend vertically and release the power bank into the target zone.	AUTOMATE pre-flagged verifier timing differs 4.2 s (corrected)	not signed off

Parts seen (estimated): power bank (~150 g, 105×66×15 mm)

Layout estimated from video



UR3e reach 500 mm needed reach 429 mm pick place

Bench 1.1 × 0.75 m; scale confidence 0.8. Scale is anchored by the MagSafe power bank (105x66 mm), a Microsoft Surface laptop on the right back corner (~300x220 mm), and the operator's shoulder width (~0.45 m) and hand span. The working end of the wooden dining table is approximately 1.1 m along the operator's front edge by 0.75 m deep up to the overhead tripod. Positions are the model's estimate from the three views, not measurements.

Robot

UR3e (Universal Robots) · passes the checks

Payload 3 kg · needs 1.15 kg ✓

Reach 500 mm · needs 429 mm ✓ · estimated layout

Price USD 23k–28k (indicative)

Arm	Payload	Reach	Price*	Verdict
UR3e Universal Robots	3 kg ✓	500 mm ✓	USD 23k–28k	**chosen**
xArm 6 UFACTORY	5 kg ✓	700 mm ✓	USD 10k–14k	also adequate; a larger arm than this job needs
UR5e Universal Robots	5 kg ✓	850 mm ✓	USD 35k–42k	also adequate; a larger arm than this job needs
Franka FR3 Franka Robotics	3 kg ✓	855 mm ✓	USD 20k–30k	also adequate; a larger arm than this job needs
Doosan M0609 Doosan Robotics	6 kg ✓	900 mm ✓	USD 30k–40k	also adequate; a larger arm than this job needs
UR10e Universal Robots	12.5 kg ✓	1,300 mm ✓	USD 50k–60k	also adequate; a larger arm than this job needs

Needed payload 1.15 kg: the heaviest of the parts in the automatable steps is power bank at about 150 g plus a 1 kg allowance for the gripper and fingers, assuming the rated payload is at the tool flange and excludes the gripper. Needed reach 429 mm: the farthest pick or place of the automatable steps from the proposed base, kept within 85%

of the rated reach; the layout is estimated from video, so treat it as approximate. UR3e (3 kg, 500 mm) is the smallest arm in the catalogue that passes every check, with 1.85 kg of payload margin; also adequate: xArm 6, UR5e, Franka FR3, Doosan M0609, UR10e.

Cycle time

6.5 s per cycle today over 3 cycles; automatable share 19 %.



- 1 · Pick power bank from table — 1.0 s · 15 %
- 2 · Reorient and inspect power bank — 1.9 s · 29 %
- 3 · Place power bank on table — 1.2 s · 19 %
- not covered by a step — 2.4 s · 37 %

With a cobot (estimate): robot 1.6 s on its steps (× 1.3), operator 2.9 s, side by side 5.3 s. Robot time = the operator's time on the automate steps x 1.3, a conservative factor for a cobot moving at collaborative speed; no robot motion was planned or simulated. Side by side = the longer of the robot's time and the operator's remaining time, the operator keeping the seconds no step covers.

Payback at the default inputs

Inputs	SGD 14/h · 2 × 8 h · 250 days · robot SGD 40,000 (indicative) × 3 · grant 50 % capped at SGD 30,000
Investment	SGD 120,000
Grant	− SGD 30,000
Savings a year	SGD 10,584
Payback	**≈ 102.0 months**

Investment = robot price × system multiple (a complete cell costs 3 to 5 × the robot: Automation World; Fraunhofer IPA). Grant: Singapore's Productivity Solutions Grant (Enterprise Singapore); check the current terms. Edit every input on the analysis page.

Cobot twin (simulated)

UR5e model for the UR3e, base at 0.26 × 0.67 m from the bench corner: every automatable point reachable, collision-free; robot 3.4 s on its steps, side by side 5.3 s.

Step	Point	Reachable	Error	Collision
Place power bank on table	pick	yes	0 mm	none
Place power bank on table	place	yes	0 mm	none

Parts list (indicative): UR3e cobot arm (Universal Robots) USD 23,000–USD 28,000; Two-finger parallel gripper (Robotiq 2F-85 class) with fingers USD 4,000–USD 6,500; Locating nest or presenter, custom (one per needs-fixture step) USD 800–USD 2,500; Pedestal or bench mount, cabling, control box placement USD 800–USD 2,000; hardware total USD 28,600–USD 39,000.

UR3e has no MuJoCo Menagerie model; simulated with the UR5e model, the nearest in the catalogue. Reach and timing are the UR5e's. Place power bank on table: no pick point of its own; the part is taken from the point of step s1 (Pick power bank from table).. Kinematic twin in MuJoCo with the Menagerie arm model: the base is placed where every automatable point is within 85 % of the rated reach with the most margin; the tool goes pick, lift, place with inverse kinematics (mink); each configuration and the path between them are checked for contact with the bench, the objects and the fixtures; time = the longer of joint travel at 60°/s and tool travel at 0.25 m/s, plus 0.5 s to close or open the gripper. The part is welded to the gripper, not grasped; the gripper is a 160 mm tool; nothing here is a physics or safety validation.

Engineer's notes

The proposed cobot cell would take over only the basic placement of the power bank, leaving picking preparation, mid-air reorientation, and visual inspection with the operator. With only a single 1.2-second action automated, the estimated payback period is 102 months against a net investment of SGD 90,000. The main caveat is that the project is commercially impractical in its current scope unless additional kitting and inspection tasks are automated to relieve operator workload.

Risks

- **Severe payback delay** · The 102-month payback offers minimal economic return because the cobot absorbs only 1.2 seconds of task time. Re-evaluate whether automating other kitting steps can share the equipment cost before issuing a purchase order.
- **Gripper collision on unindexed parts** · Attempting to pick a flat 15 mm profile directly from an unguided table surface risks gripper crashes or missed picks. Bench test jaw clearance and part elevation using sample parts and 3D-printed fingers prior to hardware procurement.
- **Operator pacing bottleneck** · Because manual reorientation and visual checking remain on the human operator, the robot will sit idle waiting for cycle handoffs. Measure operator cycle time variance across a full shift to see if the projected 5.27-second cycle is realistically maintainable.

Next steps

1. Fabricate a 3D-printed raised fixture prototype with finger reliefs to validate picking the 15 mm power bank with parallel jaws.
2. Evaluate whether a stationary machine vision camera and a static passive regrip stand can feasibly automate step s2.
3. Map the full kitting table workspace to verify that all tray and bin locations fit comfortably within the UR3e 500 mm reach envelope.
4. Request formal system integration and tooling quotes from Universal Robots distributors to firm up the SGD 120,000 budget.

Before committing

- Can the visual inspection step be codified into a pass/fail camera inspection rather than subjective human review?
- Can power banks be delivered to the station pre-oriented in slotted magazines rather than lying loose on the table?
- Is there budget and justification to automate the remaining parts in bins A, B, and C to dramatically improve payback?

Written by gemini-3.8-flash from the analysis, not from the video; the verdicts were not changed.

Analysis a-20261017-052107-9749 · model gemini-4-argon · 32,111 prompt and 17,269 output tokens · 4 min 34 s · Every verdict, time and reason comes from the model's stored answer; the robot choice, cycle and payback arithmetic are the app's code. Printed Oct 17, 2026, 07:55 PM.